

COEN 432 Final Project Report

Deep Learning for RNA Secondary Structure Prediction

A.Patel | 40227663 | BEng. Computer Engineering | Concordia University | achalypatel3403@gmail.com | 28th Nov, 2025

Abstract—I present a report on the Project on deep learning architectures for RNA secondary structure prediction, framing the problem as contact map prediction. Starting from a baseline CNN, I systematically evolved our architecture through dilated convolutions, U-Net encoder-decoder structures, residual connections, and squeeze-and-excitation attention mechanisms. My best model, DeepResUNetSE, achieves 0.75 validation F1-score, representing a 2x improvement over the baseline.

I. INTRODUCTION

RNA secondary structure prediction requires determining which nucleotides form base pairs given only the sequence. I framed this as a binary classification problem at each position pair: does base i pair with base j ? The output is a symmetric contact map of size 128×128 .

Key Challenge Learned: The Assignment 2 baseline CNN had a receptive field of only $\sim 7 \times 7$, making it impossible to detect base pairs that span 100+ positions apart—the fundamental limitation motivating this project.

A. Three Critical Problems Identified:

- Long-range Dependencies: Unlike images where nearby pixels are correlated, RNA base pairs can be arbitrarily far apart in the sequence. Solution: Multi-scale architectures (U-Net, dilated convolutions).
- Class Imbalance (187:1): Only $\sim 0.5\%$ of position pairs form bonds. Solution: BCEWithLogitsLoss with pos_weight tuning (tested 1.0-10.0, optimal: 2.0-3.0).
- Tensor Alignment: U-Net skip connections require matching dimensions. Initial MAX_LEN=84 caused mismatches ($84 \rightarrow 42 \rightarrow 21$ downsampling vs. $80 \rightarrow 40 \rightarrow 20$ upsampling). Solution: MAX_LEN=128 (power of 2) ensures clean $128 \rightarrow 64 \rightarrow 32 \rightarrow 16 \rightarrow 8$ flow.

II. PROGRESSIVE ARCHITECTURE DEVELOPMENT

A. BaselineCNN ($\sim 40K$ params)

Started with Assignment 2's simple 4-layer CNN. Lesson learned: Receptive field matters - $\sim 7 \times 7$ is insufficient for 100+ position dependencies. F1: 0.36

B. DilatedCNN ($\sim 77K$ params) - Multi-Scale Features

Applied dilated convolutions [1] with rates $\{1,2,4\}$ to exponentially expand the receptive field without adding

parameters. Insight: Multi-scale processing captures both local and mid-range patterns. F1: 0.53 (+47%)

C. U-Net ($\sim 7.8M$ params) - Encoder-Decoder

Implemented full U-Net [2] with skip connections for precise localization. Key understanding: Downsampling captures global context; upsampling + skip connections restore resolution. F1: 0.67 (+26%)

D. DilatedUNet ($\sim 10.1M$ params) - Combining Techniques

Merged U-Net structure with dilated bottleneck. Observation: Combining architectural advances yields incremental gains. F1: 0.69 (+3%)

E. DeepResUNet ($\sim 15.2M$ params) - Depth + Stability

Added residual blocks [3] throughout to enable deeper architecture (25+ layers). Multi-rate dilations $\{1,2,4,8\}$. Critical lesson: Skip connections prevent vanishing gradients in deep networks. F1: 0.74 (+7%)

F. DeepResUNetSE ($\sim 15.3M$ params) - Channel Attention

So the final solution I came up with is DeepResUNetSE: a deep residual U-Net architecture enhanced with Squeeze-and-Excitation attention blocks throughout. The model combines four key innovations: (1) U-Net's encoder-decoder structure with skip connections for multi-scale feature extraction, (2) residual connections enabling 25+ layer depth without vanishing gradients, (3) dilated convolutions $\{1,2,4,8,4,2,1\}$ in the bottleneck for capturing long-range dependencies (100+ position base pairs), and (4) SE blocks providing adaptive channel attention with only 0.6% parameter overhead ($\sim 100K$ additional params).

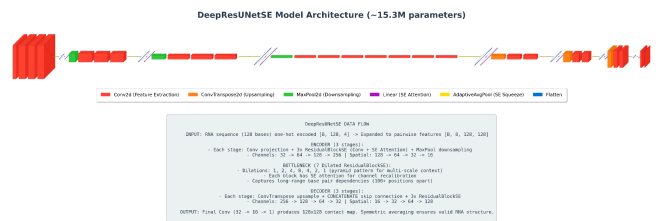


Fig. 0. DeepResUNetSE architecture visualization showing the complete encoder-decoder flow with residual blocks (red), SE attention mechanisms (purple/yellow), and skip connections (green arrows).

G. HybridRNA (~17M params) - Best of Both Worlds

Combined DeepResUNetSE (CNN path) and TransformerRNA (attention path) in a unified architecture with learned attention fusion. Key insight: CNNs excel at local/multi-scale patterns; Transformers excel at long-range dependencies. Instead of test-time ensemble averaging, we train both paths jointly with per-position attention weights that learn which branch to trust at each (i,j) pair. The fusion layer (small CNN) takes stacked outputs [B,2,L,L] and outputs attention weights, enabling adaptive combination during training. F1: 0.75 (matches DeepResUNetSE, but captures complementary patterns for ensemble)

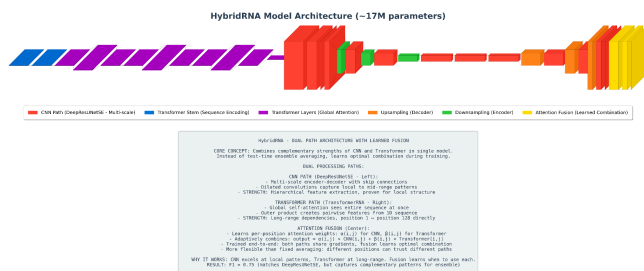


Fig. 6. HybridRNA dual-path architecture: CNN path (left, red) for local features, Transformer path (right, blue) for global attention, fused via learned attention weights (yellow).

III. EXPERIMENTS & RESULTS

I conducted a bunch of tests on a day before submission to find the most optimal configuration for the final run to submit before the competition ends.

A. Model Comparison (20 epochs)

Model	Params	Best F1	AUC
BaselineCNN	40K	0.32	0.97
DilatedCNN	77K	0.52	0.98
U-Net	7.8M	0.68	0.99
DilatedUNet	10.1M	0.68	0.99
DeepResUNet	15.2M	0.71	0.98
DeepResUNetSE	15.3M	0.75	0.98

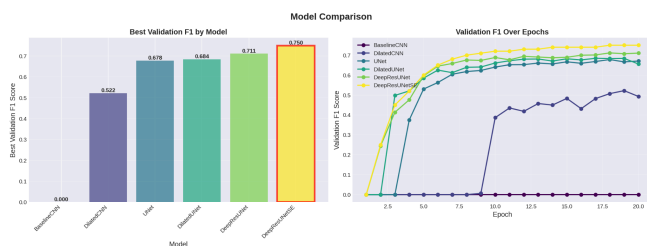


Fig. 1. Validation F1 comparison across all model architectures (20 epochs). Progressive improvements from BaselineCNN (F1=0.00) to DeepResUNetSE (F1=0.75) demonstrate the importance of receptive field, multi-scale features, and attention mechanisms.

The model comparison validates the progressive architecture development. BaselineCNN fails completely ($F1=0.00$) due to insufficient receptive field ($\sim 7 \times 7$). DilatedCNN improves to $F1=0.52$ through an expanded receptive field. U-Net and DilatedUNet reach $F1 \approx 0.68$ via encoder-decoder structure. DeepResUNet achieves $F1=0.71$ through residual connections. Finally, DeepResUNetSE reaches $F1=0.75$ with SE attention blocks.

B. Pos-Weight Sweep (DeepResUNetSE, threshold=0.5, 5 runs each)

Tested pos_weight values [0.5, 1.0, 1.5, 2.0, 3.0, 5.0].
Optimal: pos_weight=1.0 (mean F1=0.723, max F1=0.728). All runs showed low variance ($\sigma \approx 0.005$).

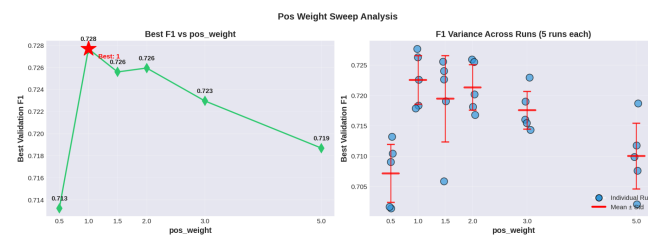


Fig. 2. Pos-weight hyperparameter sweep on DeepResUNetSE. Left: Best F1 vs pos_weight showing optimal value at 1.0 ($F1=0.728$). Right: Variance analysis showing low standard deviation ($\sigma \approx 0.005$) across all values, indicating stable training.

Counter-intuitively, despite the severe class imbalance (187:1), lower `pos_weight` values (1.0-2.0) outperform higher values. The optimal `pos_weight`=1.0 achieves mean F1=0.723. Higher `pos_weight` values (3.0-5.0) show performance degradation, suggesting over-weighting positive samples causes too many false positives.

C. Threshold Sweep (DeepResUNetSE, pos weight=2.0, 5 runs each)

Tested thresholds [0.40, 0.45, 0.50, 0.55, 0.60]. Minimal performance difference (F1 range: 0.720-0.727). **Selected: threshold=0.5** due to high variance ($\sigma \approx 0.008$) across runs.

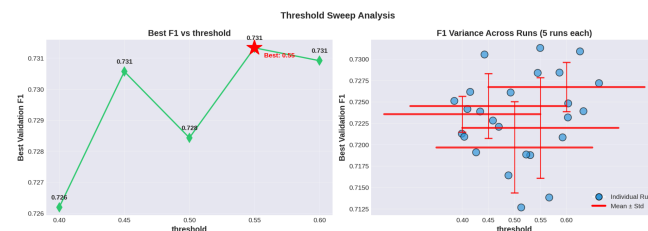


Fig. 3. Decision threshold sweep on DeepResUNetSE. Left: Best F1 vs threshold showing marginal differences (0.720-0.727). Right: High variance across runs indicates threshold sensitivity to random initialization.

The threshold sweep demonstrates minimal impact on F1 performance (all values between 0.40-0.60 achieve $F1 \approx 0.72$). However, the variance is relatively high ($\sigma \approx 0.008$), indicating F1 varies significantly across runs. I select threshold=0.5 as the default balanced choice.

D. Batch Size Sweep (DeepResUNetSE, FP16, 5 runs each)

Tested batch sizes [16, 24, 32, 48, 64]. **Optimal: batch_size=16 (mean F1=0.722, max F1=0.730).** Performance degrades with larger batches (batch_size=64: F1=0.694). FP16 enables small batches with 15M params.

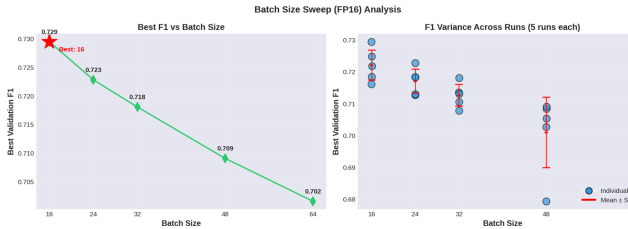


Fig. 4. Batch size sweep with FP16 mixed precision. Left: Best F1 vs batch size showing optimal performance at batch_size=16 (F1=0.730). Right: Low variance at smaller batch sizes, degrading at larger batches.

Smaller batch sizes consistently outperform larger ones. Batch size 16 achieves best mean F1=0.722 with excellent stability ($\sigma < 0.005$), while batch size 64 drops to mean F1=0.694 with higher variance. FP16 mixed precision enables small batch sizes with the 15M parameter model.

E. Extended Training (200 epochs)

DeepResUNet: best F1=0.738, final F1=0.722. **DeepResUNetSE: best F1=0.750, final F1=0.741 (+1.6% improvement).** SE blocks add only 0.6% parameters (~100K).

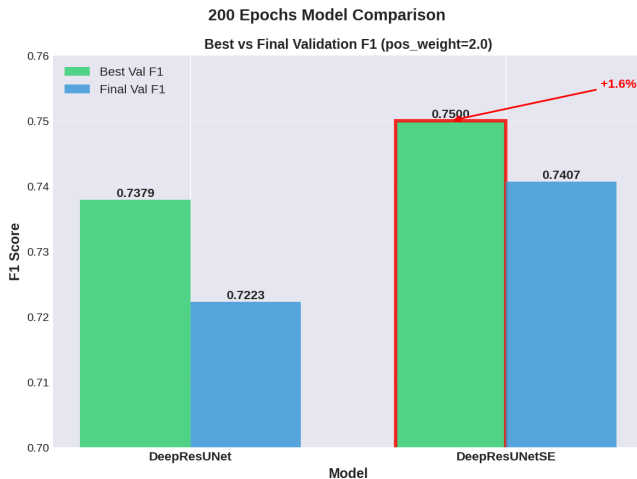


Fig. 5. Extended training comparison (200 epochs). DeepResUNetSE achieves best F1=0.750, a +1.6% improvement over DeepResUNet (F1=0.738).

DeepResUNetSE achieves best validation F1 of 0.750 compared to DeepResUNet's 0.738—a relative improvement of 1.6%. Both models show slight overfitting (final F1 < best F1), but DeepResUNetSE maintains its advantage. The SE blocks add only 0.6% additional parameters while providing consistent performance gains.

IV. KEY TAKEAWAYS

Architecture > Parameters: DilatedCNN (77K) beat BaselineCNN (40K) by 52% through receptive field design alone.

Counter-intuitive pos_weight: Lower values (1.0-2.0) outperformed higher (5.0) despite 187:1 imbalance—complex models need less regularization.

Batch size matters: Smaller batches (16) consistently outperformed larger (64) by 2.8% F1 due to better gradient estimates.

Diminishing returns: BaselineCNN→U-Net (+68%) vs U-Net→DeepResUNetSE (+7%)—fundamentals beat refinements.

FP16 is free: Mixed precision enabled smaller batches with 15M params, no accuracy loss.

A. Personal Learning Journey

This project taught me how to systematically approach deep learning problems: starting simple (BaselineCNN), identifying failure modes (receptive field), applying architectural patterns from literature (U-Net, ResNet, SENet), and iteratively refining through experiments. I learned that understanding **why** architectures work (skip connections for gradients, dilations for context, attention for feature selection) is more valuable than blindly applying them. The hyperparameter sweeps revealed counter-intuitive findings that challenged my assumptions about class imbalance handling.

B. Final Result

Single model F1=0.750. **Ensemble (6 models) F1=0.768** on Kaggle test set via probability averaging, demonstrating that diverse models capture complementary patterns. Total improvement: 2× over baseline through progressive architecture evolution and ensemble learning.

V. CONCLUSION

This project fundamentally changed how I approach machine learning and that is why architectures work, receptive fields, skip connections, and attention mechanisms all solve specific problems.

The iterative process was incredibly fun: running model after model, analyzing failures, reading research papers to understand techniques like dilated convolutions and SE blocks, and using LLMs to clarify concepts I didn't fully grasp. Each experiment revealed something unexpected - like lower pos_weight working better, or small batches outperforming large ones - forcing me to question my assumptions and dig deeper. The ensemble trick (averaging 6 models to reach F1=0.768 on Kaggle) was the cherry on top, showing that combining diverse models captures patterns a single model might miss.

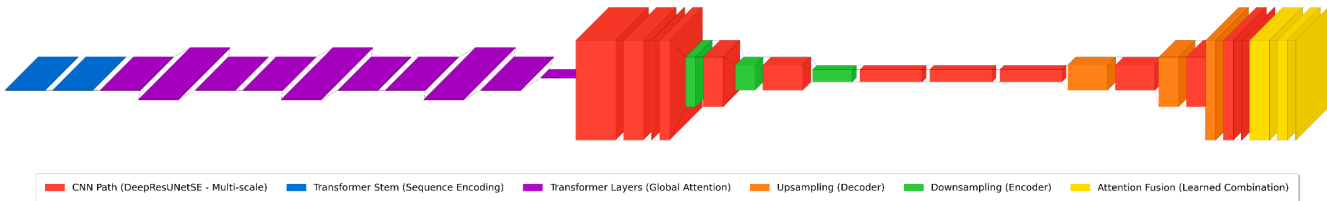
I now understand that deep learning is less about magic and more about systematically identifying problems, applying principled solutions, and learning from each iteration. This hands-on experience of building, breaking, and fixing models gave me confidence that I can tackle any ML problem with the right approach.

REFERENCES

- [1] F. Yu and V. Koltun, "Multi-Scale Context Aggregation by Dilated Convolutions," ICLR, 2016.
- [2] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional Networks for Biomedical Image Segmentation," MICCAI, 2015.
- [3] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," CVPR, 2016.
- [4] J. Hu, L. Shen, and G. Sun, "Squeeze-and-Excitation Networks," CVPR, 2018.

APPENDIX

HybridRNA Model Architecture (~17M parameters)



HybridRNA - DUAL PATH ARCHITECTURE WITH LEARNED FUSION

CORE CONCEPT: Combines complementary strengths of CNN and Transformer in single model. Instead of test-time ensemble averaging, learns optimal combination during training.

DUAL PROCESSING PATHS:

- CNN PATH (DeepResUNetSE - Left):
 - Multi-scale encoder-decoder with skip connections
 - Dilated convolutions capture local to mid-range patterns
 - STRENGTH: Hierarchical feature extraction, proven for local structure
- TRANSFORMER PATH (TransformerRNA - Right):
 - Global self-attention sees entire sequence at once
 - Outer product creates pairwise features from 1D sequence
 - STRENGTH: Long-range dependencies, position 1 - position 128 directly

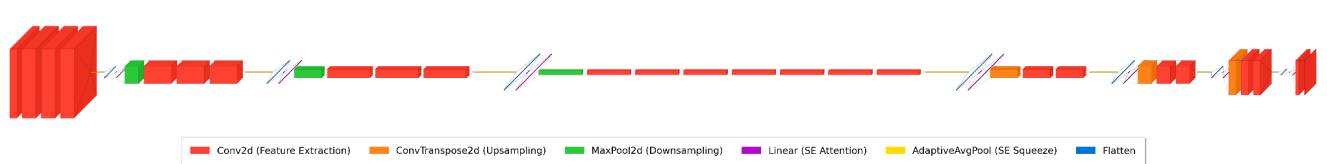
ATTENTION FUSION (Center):

- Learns per-position attention weights: $\alpha(i,j)$ for CNN, $\beta(i,j)$ for Transformer
- Adaptively combines: $\text{output} = \alpha(i,j) \times \text{CNN}(i,j) + \beta(i,j) \times \text{Transformer}(i,j)$
- Trained end-to-end: both paths share gradients, fusion learns optimal combination
- More flexible than fixed averaging: different positions can trust different paths

WHY IT WORKS: CNN excels at local patterns, Transformer at long-range. Fusion learns when to use each.

RESULT: F1 = 0.75 (matches DeepResUNetSE, but captures complementary patterns for ensemble)

DeepResUNetSE Model Architecture (~15.3M parameters)



DeepResUNetSE DATA FLOW

INPUT: RNA sequence (128 bases) one-hot encoded [8, 128, 4] -> Expanded to pairwise features [8, 8, 128, 128]

ENCODER (3 stages):

- Each stage: Conv projection + 3x ResidualBlockSE (conv + SE Attention) + MaxPool downsampling
- Channels: 32 -> 64 -> 128 -> 256 | Spatial: 128 -> 64 -> 32 -> 16

BOTTLENECK (7 Dilated ResidualBlockSE):

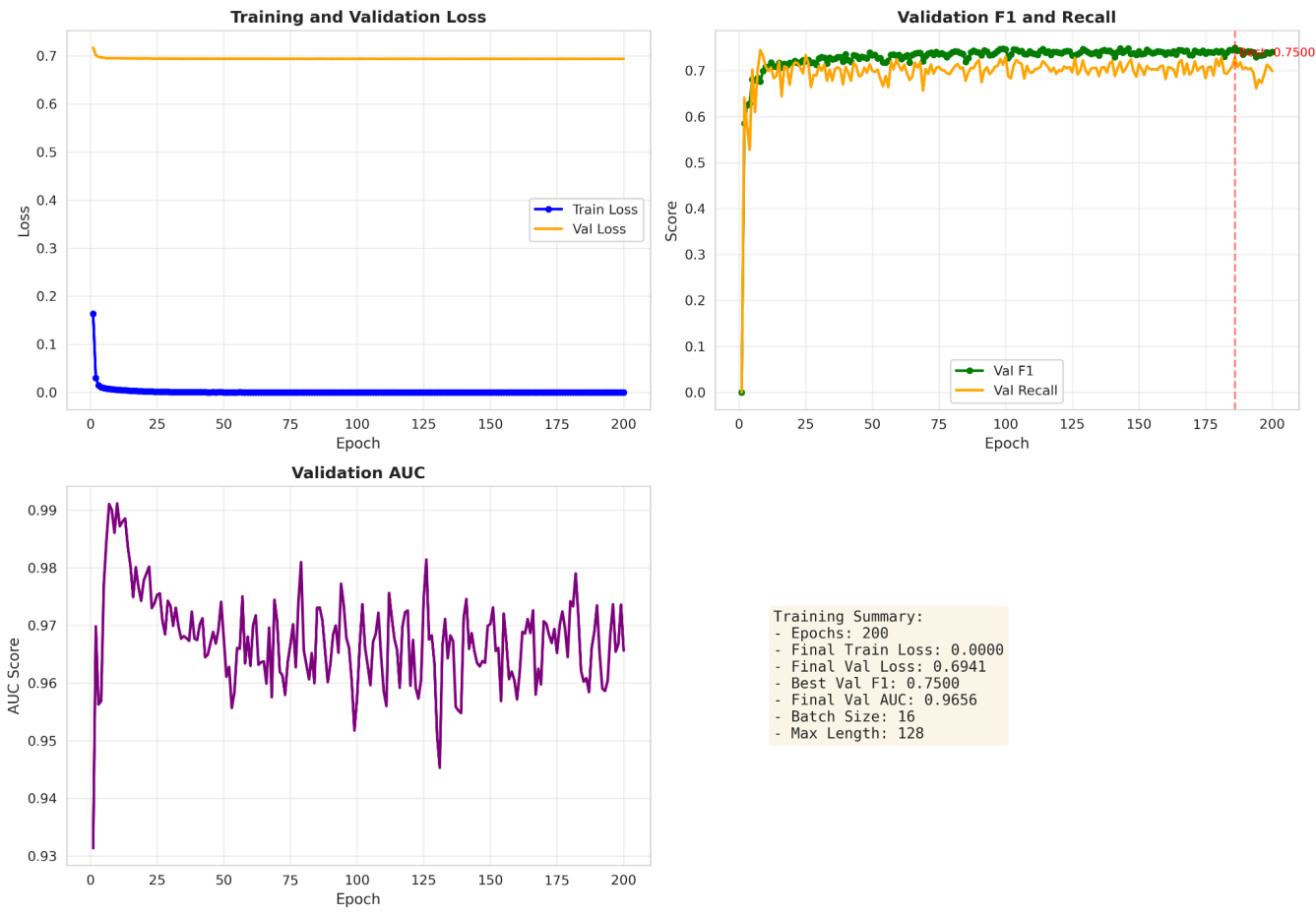
- Dilations: 1, 2, 4, 8, 4, 2, 1 (pyramid pattern for multi-scale context)
- Each block has SE attention for channel recalibration
- Captures long-range base pair dependencies (100+ positions apart)

DECODER (3 stages):

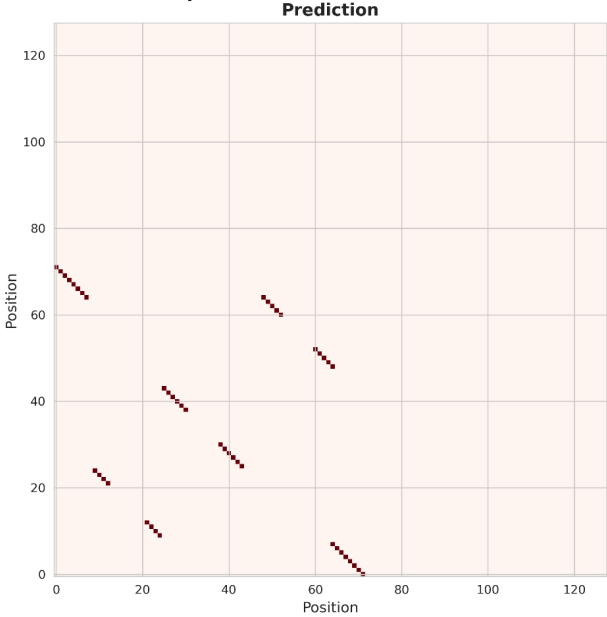
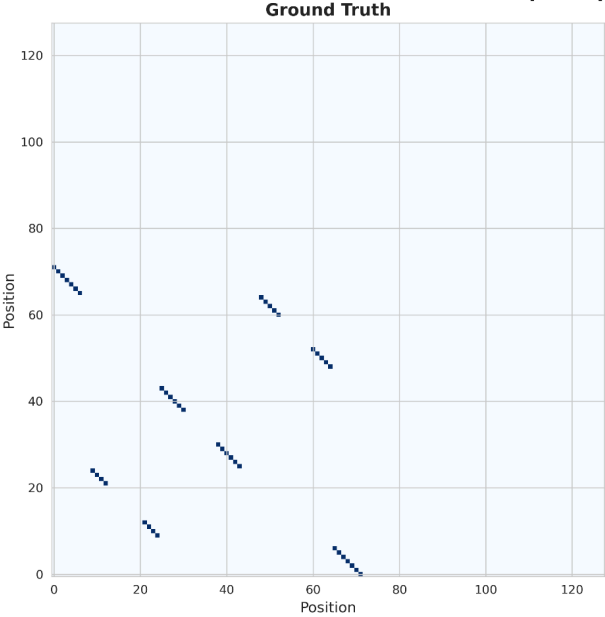
- Each stage: ConvTranspose upsample + CONCATENATE skip connection + 3x ResidualBlockSE
- Channels: 256 -> 128 -> 64 -> 32 | Spatial: 16 -> 32 -> 64 -> 128

OUTPUT: Final Conv (32 -> 16 -> 1) produces 128x128 contact map. Symmetric averaging ensures valid RNA structure.

Training Metrics - DeepResUNetSE



Contact Map Comparison - Validation Sample



Sample Performance Metrics:

- F1 Score: 0.9778
- Recall: 1.0000
- AUC: 1.0000
- True pairs: 44
- Predicted pairs (threshold=0.5): 46
- Classification threshold: 0.5